KAIZEN MATHS

Lesson starter

Partial Fractions: Decomposition

Improper Fractions

Name _____

S1

Express in partial fractions:

$\frac{-3x^2+13x-10}{x(x-2)}$

S2

Express in partial fractions:

$\frac{-2x^2+7x-7}{(x-1)^2}$

S3

Express in partial fractions:

$\frac{-2x^2-3x+12}{(x+4)x}$

S4

Express in partial fractions:

$\frac{-4x^2-2x-6}{x^2}$

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ALG-36C3-L3 · S

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Exit ticket

Partial Fractions: Decomposition

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E1

Express in partial fractions:

$\frac{-3x^2-16x-26}{(x+2)(x+4)}$

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Express in partial fractions:

$\frac{-2x^2+13x-27}{(x-3)^2}$

E3

Express in partial fractions:

$\frac{4x^2-10x-38}{(x+3)(x-4)}$

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Teacher answers & working

Partial Fractions: Decomposition · Improper Fractions

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S1

Express in partial fractions:
$\frac{-3x^2+13x-10}{x(x-2)}$

Answer:

$(-3) + \frac{5}{x} + \frac{2}{(x-2)}$

Working:

The numerator has the same degree as the denominator, so divide first.

$-3x^2 + 13x - 10 = -3(x(x - 2)) + (7x - 10)$

$\frac{-3x^2+13x-10}{x(x-2)} = -3 + \frac{7x-10}{x(x-2)}$

Now decompose the proper fraction.

$\frac{7x-10}{x(x-2)} = \frac{A}{x} + \frac{B}{(x-2)}$

$7x - 10 = A(x - 2) + Bx$

Using x = 0 gives A = 5.

Using x = 2 gives B = 2.

Therefore $\frac{-3x^2+13x-10}{x(x-2)} = (-3) + \frac{5}{x} + \frac{2}{(x-2)}$.

S2

Express in partial fractions:
$\frac{-2x^2+7x-7}{(x-1)^2}$

Answer:

$(-2) + \frac{3}{(x-1)} - \frac{2}{(x-1)^2}$

Working:

The fraction is improper, so divide first.

$-2x^2 + 7x - 7 = -2((x - 1)^2) + (3x - 5)$

$\frac{-2x^2+7x-7}{(x-1)^2} = -2 + \frac{3x-5}{(x-1)^2}$

Use the repeated-factor form for the proper fraction.

$\frac{3x-5}{(x-1)^2} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2}$

$3x - 5 = A(x - 1) + B$

Comparing x-coefficients gives A = 3.

Comparing constants gives B = -2.

Therefore $\frac{-2x^2+7x-7}{(x-1)^2} = (-2) + \frac{3}{(x-1)} - \frac{2}{(x-1)^2}$.

Teacher answers & working

Partial Fractions: Decomposition · Improper Fractions

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S3

Express in partial fractions:
$\frac{-2x^2-3x+12}{(x+4)x}$

Answer:

$(-2) + \frac{2}{(x+4)} + \frac{3}{x}$

Working:

The numerator has the same degree as the denominator, so divide first.

$-2x^2 - 3x + 12 = -2((x + 4)x) + (5x + 12)$

$\frac{-2x^2-3x+12}{(x+4)x} = -2 + \frac{5x+12}{(x+4)x}$

Now decompose the proper fraction.

$\frac{5x+12}{(x+4)x} = \frac{A}{(x+4)} + \frac{B}{x}$

$5x + 12 = Ax + B(x + 4)$

Using x = -4 gives A = 2.

Using x = 0 gives B = 3.

Therefore $\frac{-2x^2-3x+12}{(x+4)x}$ = $(-2) + \frac{2}{(x+4)} + \frac{3}{x}$.

S4

Express in partial fractions:
$\frac{-4x^2-2x-6}{x^2}$

Answer:

$(-4) + \frac{-2}{x} - \frac{6}{x^2}$

Working:

The fraction is improper, so divide first.

$-4x^2 - 2x - 6 = -4(x^2) + (-2x - 6)$

$\frac{-4x^2-2x-6}{x^2} = -4 + \frac{-2x-6}{x^2}$

Use the repeated-factor form for the proper fraction.

$\frac{-2x-6}{x^2} = \frac{A}{x} + \frac{B}{x^2}$

$-2x - 6 = Ax + B$

Comparing x-coefficients gives A = -2.

Comparing constants gives B = -6.

Therefore $\frac{-4x^2-2x-6}{x^2}$ = $(-4) + \frac{-2}{x} - \frac{6}{x^2}$.

Teacher answers & working

Partial Fractions: Decomposition · Improper Fractions

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E1

Express in partial fractions:
$\frac{-3x^2-16x-26}{(x+2)(x+4)}$

Answer:

$(-3) + \frac{-3}{(x+2)} + \frac{5}{(x+4)}$

Working:

The numerator has the same degree as the denominator, so divide first.

$-3x^2 - 16x - 26 = -3((x + 2)(x + 4)) + (2x - 2)$

$\frac{-3x^2-16x-26}{(x+2)(x+4)} = -3 + \frac{2x-2}{(x+2)(x+4)}$

Now decompose the proper fraction.

$\frac{2x-2}{(x+2)(x+4)} = \frac{A}{(x+2)} + \frac{B}{(x+4)}$

$2x - 2 = A(x + 4) + B(x + 2)$

Using x = -2 gives A = -3.

Using x = -4 gives B = 5.

Therefore $\frac{-3x^2-16x-26}{(x+2)(x+4)}$ = $(-3) + \frac{-3}{(x+2)} + \frac{5}{(x+4)}$.

E2

Express in partial fractions:
$\frac{-2x^2+13x-27}{(x-3)^2}$

Answer:

$(-2) + \frac{1}{(x-3)} - \frac{6}{(x-3)^2}$

Working:

The fraction is improper, so divide first.

$-2x^2 + 13x - 27 = -2((x - 3)^2) + (x - 9)$

$\frac{-2x^2+13x-27}{(x-3)^2} = -2 + \frac{x-9}{(x-3)^2}$

Use the repeated-factor form for the proper fraction.

$\frac{x-9}{(x-3)^2} = \frac{A}{(x-3)} + \frac{B}{(x-3)^2}$

$x - 9 = A(x - 3) + B$

Comparing x-coefficients gives A = 1.

Comparing constants gives B = -6.

Therefore $\frac{-2x^2+13x-27}{(x-3)^2}$ = $(-2) + \frac{1}{(x-3)} - \frac{6}{(x-3)^2}$.

Teacher answers & working

Partial Fractions: Decomposition · Improper Fractions

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E3

Express in partial fractions:
$\frac{4x^2-10x-38}{(x+3)(x-4)}$

Answer:

$4 + \frac{-4}{(x+3)} - \frac{2}{(x-4)}$

Working:

The numerator has the same degree as the denominator, so divide first.

$4x^2 - 10x - 38 = 4((x + 3)(x - 4)) + (-6x + 10)$

$\frac{4x^2-10x-38}{(x+3)(x-4)} = 4 + \frac{-6x+10}{(x+3)(x-4)}$

Now decompose the proper fraction.

$\frac{-6x+10}{(x+3)(x-4)} = \frac{A}{(x+3)} + \frac{B}{(x-4)}$

$-6x + 10 = A(x - 4) + B(x + 3)$

Using x = -3 gives A = -4.

Using x = 4 gives B = -2.

Therefore $\frac{4x^2-10x-38}{(x+3)(x-4)} = 4 + \frac{-4}{(x+3)} - \frac{2}{(x-4)}$.